Astronomy Key

CSE 2025

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36 Questions
Total: 50 Points

Assume all questions are worth 1 point each unless otherwise noted (Q16, Q33, Q34, Q35, Q35)

1. B
2. E
3. Main Sequence
4. Pauli Exclusion Principle
5. B
6. A
7. A
8. E
9. D
10. C
11. Accretion (accept a synonym or similar word)
12. 5 billion years (accept any answer that's *less* than 1 billion years off)
13. C
14. C
15. D
16. 1 pt each for Spiral, Elliptical, and Irregular
17. A
18. C
19. D
20. C
21. B
22. A
23. D
24. B
25. E
26. B
27. D
28. C
29. D

30. D
31. C
32. B
33. (4 pts) x must be ⅓ so the meters^3 of the term in the parenthesis can be reduced to meters to match the radius in meters on the other side of the equation (Give 2 pts if the correct answer is written without supporting work. Give 1 point for partially correct analysis of the units. Do *not* give any credit if the answer is incorrect, even if it's close, and there's no supporting work)
34. (4 pts) 11.11 parsecs (Give partial credit based on correctness of work. Be aware a common mistake might be forgetting to halve the angle. Correct work should involve the d=1/p formula)
35. (4 pts) 1.50 W/m^2 (A correct answer should utilize the inverse-square law. Accept answers from 1.45 to 1.55. Give partial credit based on correctness of work)
36. (4 pts)

$v_e = \sqrt{2GM/R}$

$v_e = c$

$c^2 = 2GM/R$

Final Answer: $R = 2GM/c^2$

Substitution can be done at any point. Give partial credit based on closeness.